

Analyzing E-Learning Platform Purchases using MySQL

Project Overview and Objective

This project focuses on analyzing **e-learning platform purchase data using MySQL**. The database contains information about learners, courses, and purchase transactions. The objective is to use SQL techniques to analyze **learner spending, purchasing behavior, course popularity, sales trends, and category performance**, and to derive meaningful insights that can help the management team make better business and marketing decisions.

Problem Statement

An online learning platform sells various digital courses to learners across different countries. Each learner can purchase multiple courses, and each course belongs to a specific category. The management team wants to analyze the purchase data to understand **sales trends, learner behavior, and popular course categories**. The task is to design and query a relational database using MySQL and generate meaningful insights that can support better business and marketing decisions.

Attribute (Column /Features) Details:

Learners Table

Attribute Name	Data Type	Description
learner_id	INT (Primary Key)	Unique identifier for each learner
Full_name	VARCHAR(100)	Name of the learner
country	VARCHAR(100)	Country of residence of the learner

Courses Table

Attribute Name	Data Type	Description
course_id	INT (Primary Key)	Unique identifier for each course
course_name	VARCHAR(100)	Title/Name of the course
category	VARCHAR(100)	Category of the course
unit_price	DECIMAL(10,2)	Price of one course

Purchase Table

Attribute Name	Data Type	Description
purchase_id	INT (Primary Key)	Unique identifier for each purchase
learner_id	INT (Foreign key)	Identifies the learner who made the purchase
course_id	INT (Foreign key)	Identifies the purchased course
quantity	INT	Number of courses purchased
purchase_date	DATE	Date on which the purchase was made

Tools & Technologies

Tools and Technologies Used

- **MySQL** – Used for database creation, table design, data insertion, querying, and analysis.
- **SQL** – Used for joins, aggregations, subqueries, CTEs, CASE expressions, NULL handling, and views.
- **MySQL Workbench** – Used to create, execute, and test SQL queries and manage the database.

SQL Analysis – Questions and Results

1. Create a new database.

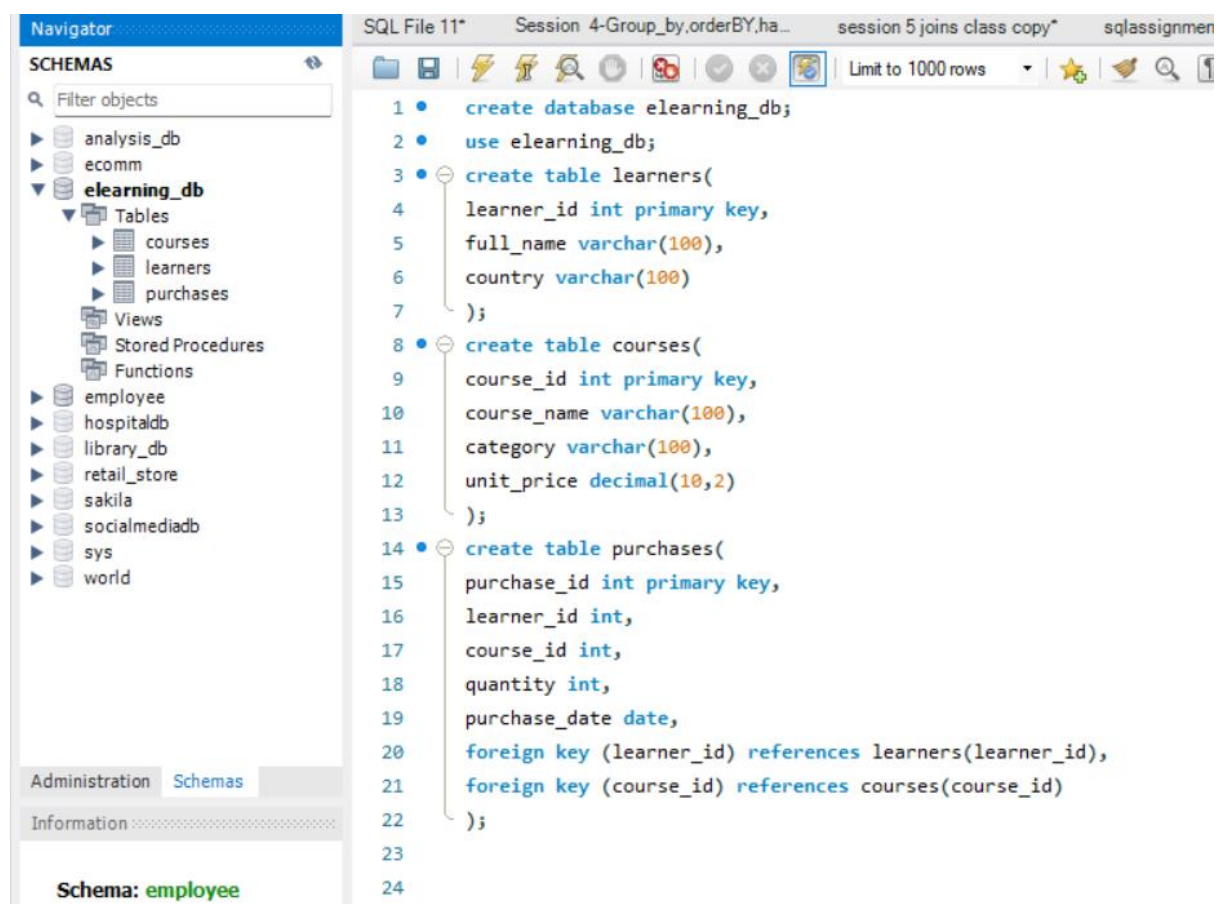
Answer: Created the elearning_db database for the e-learning purchase analysis.

2. Create the required tables with primary keys and foreign keys.

Answer: Created three tables: learners, courses, and purchases with appropriate primary and foreign key relationships.

3. Insert sample data into the tables.

Answer: Inserted sample learner, course, and purchase records into the three tables.



4. Combine learner, course, and purchase data using JOINS.

Answer: Used INNER JOIN to combine learner, course, and purchase information and calculated the total purchase amount.

```

55 • select
56     l.full_name as Learner,
57     c.course_name as Course,
58     c.category as Category,
59     p.quantity as Quantity,
60     format(p.quantity*c.unit_price,2) as Total_amount,
61     p.purchase_date as Purchase_on
62 from purchases p
63 inner join learners l
64 on l.learner_id=p.learner_id
65 inner join courses c
66 on c.course_id=p.course_id
67 order by Total_amount desc;

```

	Learner	Course	Category	Quantity	Total_amount	Purchase_on
▶	Banupriya	Advanced SQL	Advanced	1	9,000.00	2026-01-20
	Veronica	Advanced SQL	Advanced	1	9,000.00	2026-03-10
	Abinaya	Power BI Essentials	BI	1	8,000.00	2026-01-15
	Abinaya	SQL for Beginners	Beginner	1	5,000.00	2026-01-10
	John	SQL for Beginners	Beginner	1	5,000.00	2026-02-15
	Jessica	Power BI Essentials	BI	2	16,000.00	2026-02-10
	Banupriya	Python for Data Analysis	Programming	1	11,000.00	2026-02-05
	Veronica	Python for Data Analysis	Programming	1	11,000.00	2026-03-01

5. Display each learner's total spending with their country.

Answer: Calculated each learner's total spending along with their country.

```

68
69 -- Q1. Display each learner's total spending with their country.
70 • select l.full_name as Learner,l.country,format(sum(c.unit_price*p.quantity),2) as Total_Spending
71 from learners l
72 inner join purchases p
73 on l.learner_id=p.learner_id
74 inner join courses c
75 on p.course_id=c.course_id
76 group by l.learner_id,
77 l.full_name,
78 l.country
79 order by sum(c.unit_price*p.quantity) desc;

```

	Learner	country	Total_Spending
▶	Banupriya	India	20,000.00
	Veronica	UK	20,000.00
	Jessica	USA	16,000.00
	Abinaya	India	13,000.00
	John	USA	5,000.00

6. Find the top 3 most purchased courses by quantity.

Answer: Identified the top 3 courses based on purchase quantity.

```

80
81 -- Q2. Find the top 3 most purchased courses by quantity.
82 • select c.course_name as course, sum(p.quantity) as total_quantity from
83 purchases p
84 inner join courses c
85 on c.course_id=p.course_id
86 group by c.course_name, c.course_id
87 order by sum(p.quantity) desc
88 limit 3;
89

```

Result Grid

	course	total_quantity
▶	Power BI Essentials	3
	SQL for Beginners	2
	Advanced SQL	2

7. Show each category's total revenue and number of unique learners.

Answer: Calculated category-wise total revenue and unique learners.

```

88 limit 3;
89
90 -- Q3. Show each category's: Total revenue, Number of unique learners
91 • select c.category, sum(c.unit_Price*p.quantity) as Total_revenue, count(distinct l.learner_id) as unique_learners
92 from learners l
93 inner join purchases p
94 on p.learner_id=l.learner_id
95 inner join courses c
96 on c.course_id=p.course_id
97 group by c.category
98 order by sum(c.unit_Price*p.quantity) desc;
99
100

```

Result Grid

	category	Total_revenue	unique_learners
▶	BI	24000.00	2
	Programming	22000.00	2
	Advanced	18000.00	2
	Beginner	10000.00	2

8. List learners who purchased from more than one category.

Answer: Identified learners who purchased courses from multiple categories.

```

99
100 -- Q4. List learners who purchased from more than one category.
101 • select l.full_name as learner, count(distinct c.category) as category_count
102 from learners l
103 inner join purchases p
104 on p.learner_id=l.learner_id
105 inner join courses c
106 on c.course_id=p.course_id
107 group by l.learner_id,l.full_name
108 having count(distinct c.category)>1;

```

Result Grid

learner	category_count
Abinaya	2
Banupriya	2
Veronica	2

9. Identify courses never purchased.

Answer: Identified courses that have no purchase records.

```

109
110 -- Q5. Identify courses never purchased.
111 • select c.course_id,c.course_name,c.category
112 from courses c
113 left join purchases p
114 on c.course_id=p.course_id
115 where p.purchase_id is null;

```

Result Grid

course_id	course_name	category
105	Excel Fundamentals	Beginner

10. Find learners whose total spending is above the average learner spending.

Answer: Identified learners whose spending is higher than the average learner spending.

db

```

119 • select l.learner_id,l.full_name,sum(c.unit_price*p.quantity) as total_spending
120 from learners l
121 inner join purchases p
122 on p.learner_id=l.learner_id
123 inner join courses c
124 on c.course_id=p.course_id
125 group by l.learner_id,l.full_name
126 having sum(c.unit_price*p.quantity)>
127 (
128 select avg(learner_spending) from
129 (
130 select p2.learner_id,
131 sum(c2.unit_price*p2.quantity) as learner_spending
132 from purchases p2
133 join courses c2
134 on c2.course_id=p2.course_id
135 group by p2.learner_id) as spending_table);

```

Result Grid

learner_id	full_name	total_spending
2	Banupriya	20000.00
3	Jessica	16000.00
5	Veronica	20000.00

db

id int PK
int

11. Display courses whose price is higher than any course in the Beginner category.

Answer: Identified courses whose price is higher than at least one Beginner category course.

```

136
137 -- Q7. Display courses whose price is higher than any course in the 'Beginner' category.
138 • select c.course_id,c.course_name,c.category,c.unit_price
139 from courses c
140 where c.unit_price>any
141 (
142 select unit_price from courses
143 where category="Beginner");
144

```

Result Grid

course_id	course_name	category	unit_price
101	SQL for Beginners	Beginner	5000.00
102	Advanced SQL	Advanced	9000.00
103	Power BI Essentials	BI	8000.00
104	Python for Data Analysis	Programming	11000.00
NULL	NULL	NULL	NULL

courses 19 x

12. Find learners who spent more than the average spending in their country.

Answer: Identified learners whose spending is higher than the average spending in their country.

```

145 -- Q8 . Find learners who spent more than the average spending in their country.
146 • select l.learner_id,l.full_name,l.country,sum(c.unit_price * p.quantity) as total_spending
147 from learners l
148 join purchases p
149 on l.learner_id=p.learner_id
150 join courses c
151 on p.course_id=c.course_id
152 group by l.learner_id,l.full_name,l.country
153 having sum(c.unit_price*p.quantity) >
154 (
155 select avg(learner_spending)
156 from(
157 select l2.learner_id,l2.country,sum(p2.quantity*c2.unit_price) as learner_spending
158 from learners l2
159 join purchases p2
160 on l2.learner_id=p2.learner_id
161 join courses c2
162 on p2.course_id=c2.course_id
163 where l2.country=l.country
164 )

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	learner_id	full_name	country	total_spending
▶	2	Banupriya	India	20000.00
	3	Jessica	USA	16000.00

13. Use a CTE to display learners with spending above 10,000.

Answer: Used a CTE to calculate learner-wise total spending and filtered learners spending above 10,000.

```

167 --
168 -- Q9. Use a CTE to calculate total spending per learner, then:
169 -- Display learners with spending above 10,000.
170 • with learner_spending as
171 (
172 select l.learner_id,l.full_name,sum(c.unit_price*p.quantity) as total_spending
173 from learners l
174 join purchases p
175 on l.learner_id=p.learner_id
176 join courses c
177 on p.course_id=c.course_id
178 group by learner_id,l.full_name
179 )
180 select learner_id,full_name,total_spending
181 from learner_spending
182 where total_spending>10000;
183
184
185

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	learner_id	full_name	total_spending
▶	1	Abinaya	13000.00
	2	Banupriya	20000.00
	3	Jessica	16000.00
	5	Veronica	20000.00

14. Classify learners based on spending using CASE.

Answer: Classified learners as High Value, Medium Value, or Low Value based on their total spending.

```
187 • 8,000-15,000 → "Medium Value",
188 • Below 8,000 → "Low Value".*/
189 • with learner_spending as
190 (
191   select l.learner_id,l.full_name,sum(c.unit_price*p.quantity) as total_spending
192   from learners l
193   join purchases p
194   on l.learner_id=p.learner_id
195   join courses c
196   on p.course_id=c.course_id
197   group by l.learner_id,l.full_name
198 )
199   select learner_id,full_name,total_spending,
200   case
201   when total_spending>15000 then 'High Value'
202   when total_spending>=8000 then 'Medium Value'
203   else 'Low Value'
204   end as spending_category
205   from learner_spending;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	learner_id	full_name	total_spending	spending_category
▶	1	Abinaya	13000.00	Medium Value
	2	Banupriya	20000.00	High Value
	3	Jessica	16000.00	High Value
	4	John	5000.00	Low Value
	5	Veronica	20000.00	High Value

15. Handle NULL purchase counts.

Answer: Displayed all courses and replaced NULL purchase quantities with 0 using IFNULL().

```
207 -- Q11 . NULL Handling.....Display all courses and replace NULL purchase counts with 0 using: IFNULL() or COALESCE()
208 • select c.course_id,c.course_name,ifnull(sum(p.quantity),0) as purchase_count
209   from courses c
210   left join purchases p
211   on p.course_id=c.course_id
212   group by c.course_id,c.course_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	course_id	course_name	purchase_count
▶	101	SQL for Beginners	2
	102	Advanced SQL	2
	103	Power BI Essentials	3
	104	Python for Data Analysis	2
	105	Excel Fundamentals	0

Result 25 x

16. Create category_performance_view.

Answer: Created a view showing category, total revenue, number of purchases, and average revenue per purchase.


```
213
214 /* Q12 . View
215 Create a view: category_performance_view
216 • Showing:
217 • Category
218 • Total revenue
219 • Number of purchases
220 • Average revenue per purchase*/
221 • create or replace view category_performance_view as
222 select c.category,sum(c.unit_price*p.quantity) as Total_revenue,count(p.purchase_id) as Number_of_purchases,
223 round(sum(c.unit_price*p.quantity) / count(p.purchase_id),2) as Average_revenue_per_purchase
224 from courses c
225 join purchases p
226 on p.course_id=c.course_id
227 group by c.category;
228
229 • select * from category_performance_view;
230
231
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [A](#)

	category	Total_revenue	Number_of_purchases	Average_revenue_per_purchase
▶	Beginner	10000.00	2	5000.00
	BI	24000.00	2	12000.00
	Advanced	18000.00	2	9000.00
	Programming	22000.00	2	11000.00

Key Insights

- Learner spending was calculated and compared across learners
- The most purchased courses were identified
- Category-wise revenue and learner participation were analyzed
- Learners purchasing from multiple categories were identified
- Courses with no purchases were found
- High-spending learners were identified using average spending comparisons
- Course prices were compared with Beginner category prices
- Country-wise learner spending was analyzed
- Learners were classified based on their spending
- Category performance was summarized using a reusable SQL view